### 6.2.2 (ii) Construction of grouped frequency distribution table - Discontinuous series.

**Example 6.3**

Convert the given discontinuous series into a continuous series.

| Class | 0-5 | 6-11 | 12-17 | 18-23 | 24-29 |
|---|---|---|---|---|---|
| Frequency(f) | 7 | 10 | 9 | 5 | 12 |

***Solution:***

As told above, first we should fill the gap by extending the two limits of each class by half of the value of the gap. Here the gap is 1, so subtracting and adding half of the gap i.e 0.5 to the lower and the upper limit of each class makes it as a continuous series.

| Class | -0.5-5.5 | 5.5-11.5 | 11.5-17.5 | 17.5-23.5 | 23.5-29.5 |
|---|---|---|---|---|---|
| Frequency(f) | 7 | 10 | 9 | 5 | 12 |



**Try these**

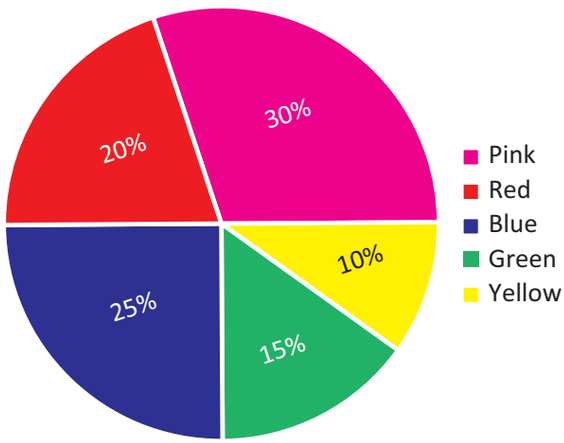
1. Prepare a frequency table for the data :
   3, 4, 2, 4, 5, 6, 1, 3, 2, 1, 5, 3, 6, 2, 1, 3, 2, 4
2. Prepare a grouped frequency table for the data :
   10, 9, 3, 29, 17, 34, 23, 20, 39, 42, 5, 12, 19, 47, 18, 19, 27, 7, 13, 40, 38, 24, 34, 15, 40

## 6.3 Graphical Representation of the Frequency Distribution for Ungrouped Data

A graphical representation is the geometrical image of a set of data. It is a mathematical picture. It enables us to think about a statistical problem in visual terms. A picture is said to be more effective than words for describing a particular thing. The graphical representation of data is more effective for understanding. In the previous classes, we have studied some graphical representations of ungrouped data such as Line graph, Bar graph, and Pictograph. Now, we are going to represent the given ungrouped data in the circular form namely the pie diagram or the pie chart.

### 6.3.1 Pie chart (or) Pie diagram

A pie chart is a circular graph which shows the total value with its components. The circle is divided into sectors and the area of the sectors is proportional to the information given. The area of a circle represents the total value and the different sectors of the circle represent the different components. In the 'pie chart' the data are mostly expressed in percentage. Each component is expressed as percentage of the total value.






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